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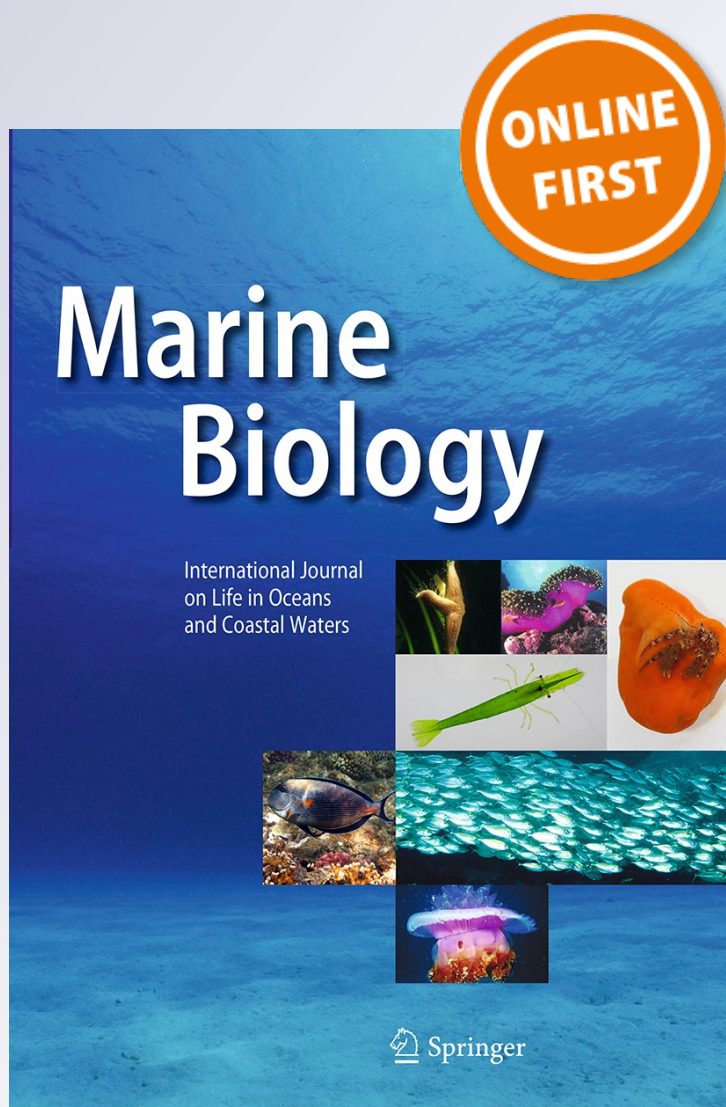
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Biogenic acidification drives density-dependent growth of a calcifying invertebrate in culture

Benjamin Mos¹  · Maria Byrne² · Kenneth L. Cowden¹ · Symon A. Dworjanyn¹

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Abstract A major factor often overlooked in the production of calcifying marine invertebrates in aquaculture is the effect of changes in the carbonate chemistry of culture water caused by organism metabolism and calcification. This study examined the effects of temperature, stocking density and seawater exchange rate on the survival and growth of the juvenile sea urchin, *Tripneustes gratilla* with particular reference to effects of these factors on the carbonate chemistry of culture water. Growth and survival of *T. gratilla* were highest at 26–28 °C. Growth rates, relative spine length, consumption rates and food conversion efficiency were reduced by up to 50 % by an increase in stocking density from 43 to 129 individuals m⁻² and a reduction in seawater exchange rate from 3.0 to 0.3 exchanges h⁻¹, but survival rates were unaffected. Analysis of the influence of seawater parameters indicated that total alkalinity (A_T), calcite saturation state (ΩCa) and dissolved CO₂ were the primary factors limiting growth at high densities and low seawater exchange rates. Uptake of bicarbonates for calcification and release of respiratory CO₂ by *T. gratilla* appeared to be driving changes in carbonate chemistry.

Our results show that reduced capacity of culture water to buffer acidification (low A_T) can be a limiting factor in intensive production of calcifying marine invertebrates. This study demonstrates the importance of considering the influence of biogenic CO₂ on seawater carbonate chemistry in aquaculture, particularly given the threat of reduced pH of source water due to ocean acidification, and also in experiments investigating the influence of anthropogenic CO₂.

Introduction

Global declines in wild-harvest fisheries and a burgeoning global population are increasing the demand for fishery products (Andrew et al. 2002; Pauly et al. 2003). Aquaculture of low trophic level species such as sea urchins is seen as a way to supply this demand without increasing pressure on wild fisheries to make high-protein aquaculture diets (Duarte et al. 2009). Development of an aquaculture industry for sea urchins has the added urgency that many wild sea urchin populations have been devastated by overfishing (Andrew et al. 2002) with substantial flow-on effects to ecosystem structure (e.g. Steneck et al. 2013). Whilst there has been considerable research on the culture of sea urchins, in 2011 the global aquaculture production of echinoderms (sea cucumbers and sea urchins) was only ~12 % of wild fishery catch (UNFAO 2011). If aquaculture is to reduce the pressure on wild-harvest fisheries, it is critical that culture techniques are optimised to increase production, efficiency and commercial viability.

Water quality is a key determinant of the performance and efficiency of aquaculture systems. Economic considerations require high stocking densities and low water exchange rates, conditions that can result in accumulation

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✉ Benjamin Mos
benjamin.mos@scu.edu.au

¹ National Marine Science Centre, Southern Cross University,
PO Box 4321, Coffs Harbour, NSW 2450, Australia

² Schools of Medical and Biological Sciences, University
of Sydney, Sydney, NSW, Australia

of metabolic wastes and/or oxygen depletion (e.g. Davidson et al. 2009; Ellis et al. 2002; Good et al. 2009; Huchette et al. 2003). Studies on sea urchins show that consumption rates, food conversion efficiency (FCE), growth rates, gonad production and survival can be negatively impacted by accumulation of ammonia, nitrite, nitrate and dissolved carbon dioxide, decreases in salinity and dissolved oxygen (DO) or sub-optimal temperatures within culture systems (Table 1). Thus far, investigations of optimal growth conditions for sea urchins have involved 11 species and focused on reproductively mature adults (Table 1). Studies of juvenile sea urchins are less common, but have shown the importance of temperature (five species), oxygen levels (one species) and stocking density (two species) (Table 1). As sea urchins appear to be sensitive to a range of water quality parameters (Table 1; Siikavuopio et al. 2007b) and juvenile sea urchins may have different tolerances to environmental conditions than adults (e.g. Siikavuopio and James 2011; Siikavuopio et al. 2007a), multifactorial studies examining the effects of water quality on performance of juvenile sea urchins within culture systems are required (Azad et al. 2010; Siikavuopio et al. 2007b).

Of particular concern for calcifying invertebrates like sea urchins is that they can alter the carbonate chemistry of their culture water, possibly compromising growth (Grosjean et al. 1998). Inorganic carbon in seawater exists as carbonate and bicarbonate ions and dissolved CO_2 , all of which covary to determine pH levels. The relationships between pH, carbonate species and the partial pressure of dissolved carbon dioxide ($p\text{CO}_2$) also depend on total alkalinity (A_T), a measure of the buffering capacity of seawater. Calcifiers may modify these parameters through release of respiratory CO_2 and uptake of carbonate ions for the construction of aragonite/calcite skeletons and regulation of internal pH. This can result in (1) reduced mineral availability requiring greater energy allocation to calcification (Doney et al. 2009; Miles et al. 2007; Stumpp et al. 2012), (2) increased $p\text{CO}_2$ causing physiological hypercapnia (Portner 2008; Widdicombe and Spicer 2008), (3) lowered pH disrupting internal acid-base balance (Miles et al. 2007; Stumpp et al. 2012) and (4) reduced A_T potentially reducing the capacity of culture water to buffer changes to carbonate chemistry parameters (Anthony et al. 2011).

Furthermore, decreased pH of source water due to increased atmospheric CO_2 (ocean acidification, Feely et al. 2004) may exacerbate the accumulation of CO_2 in culture systems with negative effects on production, as seen in the impaired growth of oysters in upwelling areas (Barton et al. 2012). The oceans are also warming—a factor of major concern because temperature is well known to play a major influence on growth of marine ectotherms. Considering the impact of these factors on invertebrate production (e.g. Table 1), it is critical to understand the relationship

between water quality and growth and survival of marine calcifying invertebrates if the aquaculture industry is to adapt.

This study investigated the impact of changes in seawater carbonate chemistry during rearing that occur in intensive aquaculture (e.g. Naylor et al. 2011). In the case of calcifying organisms, ‘acidification’ of culture water and the drawdown of A_T , generated by respiration and calcification respectively, scales with size, growth rates and stocking rates. We investigated the impacts of biogenic CO_2 , or how animals can alter their environment, in contrast to recent ocean acidification studies where conditions are altered by adding CO_2 . Specifically, we examined the effects of stocking density and seawater exchange rate on water quality and, in turn, growth and survival of the tropical sea urchin *Triplaneustes gratilla*. Although larval culture techniques (e.g. Dworjanyn and Pirozzi 2008; Mos et al. 2011), juvenile and adult diets (e.g. Dworjanyn et al. 2007; Seymour et al. 2013; Stimson et al. 2007; Vaitilingon et al. 2003) and gonad enhancement in sea cages (Juinio-Menez et al. 2008; Malay et al. 2000) have been investigated for this fast growing, high-value species (Dworjanyn et al. 2007), it is not known which water quality parameters are most important in determining the performance of juvenile *T. gratilla* in land-based culture systems. There are no other studies that we are aware of that have examined the effect of seawater exchange rates on sea urchin growth.

We investigated the effects of temperature on growth and survival of *T. gratilla* under four temperature regimes to identify optimal culture temperatures. A factorial design was then used to investigate the effect of sea urchin density and seawater exchange rate on the survival, growth, spine length, consumption rates and FCE of juvenile *T. gratilla*. Data from three density and three exchange rate treatments were assessed with respect to temperature, pH, salinity, A_T , nitrate, DO, $p\text{CO}_2$ and calcite saturation state (ΩCa) in the culture system. We also determined whether differences in seawater parameters (pH, A_T , $p\text{CO}_2$, ΩCa and nitrate concentrations) recorded among density and exchange rate treatments were attributable to the sea urchins, their faeces or food (*Sargassum* spp.), and how changes in seawater parameters were influenced by light and seawater exchange rate. Our results are placed in context with recent ocean acidification studies on sea urchin growth and projections for ocean pH decline in south-eastern Australia.

Materials and methods

Study organism

Sea urchins, *T. gratilla*, were obtained from cultures reared at the National Marine Science Centre, Southern

Table 1 Summary of the published effects of water quality parameters and stocking density on sea urchins in culture

Parameter	Species	Stage	Effect	Reference
<i>Nitrogenous wastes</i>				
Ammonia	<i>Paracentrotus lividus</i>	Adult	>1 mg L ⁻¹ reduced growth 10 mg L ⁻¹ reduced growth by ~50 %	Basuyaux and Mathieu (1999)
	<i>Strongylocentrotus droe-bachiensis</i>	Adult	>0.001 mg L ⁻¹ UIA increased mortality >0.016 mg L ⁻¹ UIA reduced gonad production No effect on feeding rate	Siikavuopio et al. (2004b)
Nitrite	<i>Paracentrotus lividus</i>	Adult	>0.5 mg L ⁻¹ reduced growth 2 mg L ⁻¹ reduced growth by up to 40 %	Basuyaux and Mathieu (1999)
	<i>Strongylocentrotus droe-bachiensis</i>	Adult	>0.5 mg L ⁻¹ reduced gonad production No effect on mortality or feeding rate	Siikavuopio et al. (2004a)
Nitrate	<i>Paracentrotus lividus</i>	Adult	>100 mg L ⁻¹ reduced growth and feeding rate	Basuyaux and Mathieu (1999)
<i>Dissolved gases</i>				
Oxygen	<i>Strongylocentrotus droe-bachiensis</i>	Juvenile	No effect of hyperoxic (>100 %) conditions on performance	Siikavuopio (2012)
		Adult	4 and 6 mg L ⁻¹ reduced growth and gonad production compared to 8 mg L ⁻¹	Siikavuopio et al. (2007b)
		Adult	No effect of hyperoxic (>100 %) conditions on performance	Siikavuopio (2012)
Carbon dioxide	<i>Lytechinus variegatus</i>	Adult	608 versus 1739 µatm Reduced growth at 1739 µatm No effect on gonad production	Challener et al. (2014)
	<i>Strongylocentrotus droe-bachiensis</i>	Adult	18.1 versus 1.1 mg L ⁻¹ Reduced gonad production, feeding rate and FCE No effect on survival	Siikavuopio et al. (2007c)
<i>Environmental</i>				
Salinity	<i>Paracentrotus lividus</i>		26, 29, 32, 35 and 38 PSU Growth rate highest at 35 PSU, reduced at all other salinities Feeding rate reduced at 26 and 38 PSU FCE lower at 26 PSU	Basuyaux et al. (1998)
Temperature	<i>Arbacia punctulata</i>	Adult	23 versus 31 °C No effect on gonad production, consumption rates or absorption efficiency	Hill and Lawrence (2006)
	<i>Eucidaris tribuloides</i>	Unknown	18 versus 27 °C Reduced feeding rates and energy absorption at 18 °C	Lares and McClintock (1991)
	<i>Evechinus chloroticus</i>	Adult	Ambient (11.3–18.2 °C) versus controlled (14.7–15.3 °C) Slightly higher gonad production in urchins with low GI in ambient regime No effect on survival	James et al. (2007)
		Adult	10, 14 and 18 °C Gonad production and consumption highest at 18 °C No effect on gonad colour	James and Heath (2008)

Cross University, Coffs Harbour, Australia (30°12.5'S, 153°16.1'E). Larval culture methods were as described by Mos et al. (2011). Larvae were induced to settle using naturally derived biofilms and *Sargassum* spp. conditioned seawater. Settled juveniles consumed the biofilm until a

test diameter (TD) of 5–10 mm when they were fed a seaweed diet (*Sargassum* spp.). Juveniles were used in experiments once they were 15–25 mm TD. Urchins were kept at 24–25 °C and not fed for 2 weeks prior to the experiments to standardise their condition.

Table 1 continued

Parameter	Species	Stage	Effect	Reference
	<i>Lytechinus variegatus</i>	Adult	16 versus 23 °C Higher absorption efficiency at 23 °C No effect on consumption rates	Klinger et al. (1986)
		Adult	23 versus 31 °C Higher gonad production and consumption at 23 °C Lower absorption efficiency except insoluble protein at 23 °C	Hill and Lawrence (2006)
		Adult	16, 22 and 28 °C Highest gonad production, growth and consumption at 22 °C Highest absorption efficiency at 28 °C	Watts et al. (2011)
	<i>Paracentrotus lividus</i>	Juvenile–Adult	‘Ambient’ versus fixed (18–20 °C) Growth rates higher under ambient regime	Fernandez and Pergent (1998)
		Adult	12, 16, 20 and 24 °C Highest gonad production at 24 °C Lower consumption rates at 12 °C	Spirlet et al. (2000)
		Adult	Gonad production negatively affected when temperature above 24–26 °C	Shpigel et al. (2004)
	<i>Strongylocentrotus droebachiensis</i>	Juvenile	5, 10 or 15 °C Survival highest at 5 °C No effect on growth	Devin et al. (2004)
		Juvenile	4.7, 9.0, 12.9, 16.0 and 19.7 °C Highest growth at 9.0 and 12.9 °C Survival highest at 12.9 and 16.0 °C	Pearce et al. (2005)
		Juvenile	6, 10, 12 and 14 °C Highest growth at 10–14 °C Highest FCE at 10 °C No difference in survival	Siikavuopio et al. (2012)
		Adult	3 versus 12 °C Higher consumption rate at 12 °C No effect on gonad production or survival	Garrido and Barber (2001)
		Adult	4, 6, 8, 10, 12 and 14 °C Gonad production highest at 12 °C (10 °C for large urchins) Highest consumption rate at 14 °C	Siikavuopio et al. (2008)
	<i>Strongylocentrotus franciscanus</i>	Juvenile	Post-settlement development fastest at 14 versus 8 and 11 °C	Miller and Emlet (1999)
		Adult	No significant difference in performance at 12 or 16 °C	McBride et al. (1997)
	<i>Strongylocentrotus intermedius</i>	Adult	Highest growth at 12 and 17 versus 8 and 22 °C No effect on feeding rate	Lawrence et al. (2009)
	<i>Strongylocentrotus purpuratus</i>	Juvenile	Post-settlement development fastest at 14 versus 8 and 11 °C	Miller and Emlet (1999)
		Adult	8.0, 12.3 and 16.2 °C Highest consumption rate and assimilation efficiency at 16 °C No effect on gonad production	Azad et al. (2011)
	<i>Tripneustes gratilla</i>	Early Juvenile	23.9, 25.1, 28.1, 30.5 and 33.4 °C Highest survival at 24–28 °C	Mos et al. (2011)

Culture method

Table 1 continued

Parameter	Species	Stage	Effect	Reference
Stocking density	<i>Evechinus chloroticus</i>	Adult	No effect of 22.4 versus 31.3 urchins m^{-2} on gonad production or survival	James (2006)
	<i>Lytechinus variegatus</i>	Juvenile/ Adult	High density (180 individuals m^{-2}) led to cannibalism, no cannibalism at 40 individuals m^{-2}	Richardson et al. (2011)
	<i>Paracentrotus lividus</i>	Adult	No effect of 20, 40, 60 or 80 urchins m^{-2} on gamete production or survival	Gago and Luís (2010)
	<i>Psammechinus miliaris</i>	Juvenile	20 versus 40 urchins 10 L^{-1} Higher growth rate at 20 urchins 10 L^{-1} No effect on survival	Kelly (2002)
	<i>Strongylocentrotus droebachiensis</i>	Juvenile	0.25, 0.5 and 1 kg m^{-2} Highest growth rate at 0.25 kg m^{-2} Reduced survival at 1 kg m^{-2} Highest injury rate at 1 kg m^{-2}	Siikavuopio and James (2011)
		Adult	3, 6, 8 and 6, 12, 14, 16 kg m^{-2} Reduced survival and gonad production with increased density No effect on feed intake	Siikavuopio et al. (2007a)
		Adult	2.5, 3.7 or 7.3 kg m^{-2} No effect on growth, FCE, gonad production, mortality or injury rate	Christiansen and Siikavuopio (2007)
	<i>Tripneustes gratilla</i>	Adult	250 versus 750 urchins m^{-2} Higher survival and gonad production at 250 urchins m^{-2} No effect on growth rate	Juinio-Menez et al. (2008)

UIA unionised ammonia, FCE food conversion efficiency

Effects of temperature on growth and survival

Experiment 1 was conducted to determine the optimal temperature for growth and survival of juvenile *T. gratilla*. Urchins were cultured under four temperature regimes (19–21, 23–25, 26–28 and 29–31 °C) for 9 weeks. Each temperature treatment had nine replicates, consisting of a 0.5 L round plastic container with a meshed window that maintained the water level at 0.4 L and containing a single randomly assigned sea urchin. Replicates were supplied with preheated seawater at an initial rate of 2.5 L h^{-1} , and as the urchins grew, this was incrementally increased to 10 L h^{-1} by the end of the experiment. Conditions were selected to minimise the effects of rearing conditions, other than temperature, on growth. Seawater for each treatment was heated using thermostat-controlled aquarium heaters in a 60 L header tank. Replicates were maintained under a 12:12 (light/dark) photoperiod supplied by ambient fluorescent lighting. Urchins were fed daily with fresh *Sargassum* spp. ad libitum. Containers were cleaned of faeces every second day. Temperatures and mortality were monitored daily in all replicates. There was a single mortality in the 29–31 °C treatment during the second week of the experiment. Mean temperatures in each treatment were 19.63 °C ($\pm 0.42 \text{ SD}$, $n = 62$), 24.1 °C ($\pm 0.75 \text{ SD}$, $n = 62$), 26.6 °C

($\pm 0.67 \text{ SD}$, $n = 62$) and 30.4 °C ($\pm 0.65 \text{ SD}$, $n = 62$) respectively.

To assess growth, test diameter (TD) and wet weight (WW) measurements were taken weekly. Urchins were weighed to the nearest 0.01 g after the removal of excess seawater by placing them on dry paper towel for 30 s. Image analysis software, ImageJ (NIH, USA) was used to measure TD from digital photographs of the aboral side of the urchins. Two perpendicular measurements at the longest axis were taken for each urchin and the average used for statistical analysis. At the beginning of the experiment, there was no significant difference in the TD (PERMANOVA, $F_{3,32} = 0.22$, $P = 0.878$; mean = $14.2 \text{ mm} \pm 0.06 \text{ SE}$) or WW ($F_{3,32} = 0.17$, $P = 0.92$; mean = $1.25 \text{ g} \pm 0.17 \text{ SE}$) of *T. gratilla* among treatments.

Effects of density and seawater exchange rate on water quality, growth and survival

Experiment 2 was conducted at ~27 °C, identified as optimal in experiment 1. A fully factorial design was used to test the effects of three stocking densities, low, medium and high (5, 10 and 15 individuals replicate $^{-1}$, respectively) and three seawater exchange rates, low, medium and high (0.3, 1.0 and 3.0 exchanges h^{-1} , respectively) on growth,

consumption rates and survival of juvenile *T. gratilla* and water quality (specifically temperature, total ammonia nitrogen (TAN), nitrite, nitrate, salinity, pH, $p\text{CO}_2$, ΩCa , A_T and DO over 10 weeks). Initial densities were equivalent to 43, 86 and 129 individuals m^{-2} (also see Fig. S1).

Each treatment had five replicates. Replicates consisted of 5 L round plastic containers (180 mm $\varnothing$ base, 200 mm H) with overflow holes that maintained the water level at 4.0 L. Replicates were randomly assigned positions in a water bath held at a depth of 100 mm to maintain stable water temperatures. Filtered (20 μm) seawater was supplied from an 800 L header tank and heated ($\sim 27^\circ\text{C}$) using a looped flow-through 3000 Watt heater (Elecro Engineering TO3 Ti). Seawater exchange rates for each replicate were regulated using irrigation drip taps. Replicates were individually aerated and maintained under a 12:12 photoperiod ('cool white' fluorescent).

Urchins were fed daily with fresh *Sargassum* spp. ad libitum. Daily temperature, pH_{NIST} and salinity measurements were recorded for each replicate and the ambient seawater supplied to the experiment using Hach[®] HQ40d multicontroller, Hach[®] PHC101 pH probe and Hach[®] CDC101 conductivity probe, respectively. A Hach[®] LDO101 probe was used to measure DO three times a week. Nitrate, nitrite and TAN concentrations were measured from water samples collected daily, pooled within treatments using equal volumes taken from each replicate, and from the ambient seawater supplied to the system (Palintest[®] photometer 7100, Ammonia AP152, Nitricol AP109 and Nitrate AP163).

Water samples (100 mL) were used to determine total alkalinity (A_T) by potentiometric titration using a Metrohm 888 Titrando and certified reference standards (Dickson et al. 2007). Water samples (100 mL) were collected weekly during the experiment, pooled from equal volumes taken from replicates in each treatment, and the ambient seawater supplied to the experiment, respectively, filtered through a 0.45 μm syringe filter, and fixed with 5 μL of saturated HgCl_2 prior to titration. Values for $p\text{CO}_2$ and ΩCa in each treatment were calculated from mean weekly temperature, pH, salinity and A_T data using CO2SYS (Pierrot et al. 2006) using the dissociation constants of Mehrbach (1973) as refitted by Dickson and Millero (1987).

Replicates were examined daily for mortalities. There was high mortality (47 %) in one replicate of the low exchange rate, high-density treatment during the first week of the experiment, with dead urchins displaying symptoms of disease. This replicate was excluded from statistical analysis.

Growth (measured as increase in TD and WW) was assessed weekly as previously described. Results were expressed as specific growth rate (SGR) and linear growth rate (LGR), respectively, calculated by:

$$\text{SGR \%} = ((\ln W_f - \ln W_i) \times 100) / (t_f - t_i)$$

$$\text{LGR} = (L_f - L_i) / (t_f - t_i)$$

where W_i and W_f were the initial and final mean WW (g), respectively, of the urchins, L_i and L_f were the initial and final mean TD (mm), respectively, of the urchins, and t_i and t_f were the initial and final time points (weeks). There was no difference in the TD ($F_{8,35} = 1.63$, $P = 0.1485$; mean = $21.6 \text{ mm} \pm 0.09 \text{ SE}$) or WW ($F_{8,35} = 0.85$, $P = 0.6391$; mean = $4.09 \text{ g} \pm 0.03 \text{ SE}$) of *T. gratilla* among treatments at the beginning of the experiment.

Spine lengths of sea urchins generally increase as test diameter increases (Tegner and Levin 1983). Therefore, we used ratios of spine length to test diameter (TD) to compare the relative spine length of *T. gratilla* among treatments. Spine length (SL) to TD ratios were calculated using the formula:

$$\text{SL} = 1/2(\text{FD}_S - \text{FD}_T) / \text{FD}_T \times 100$$

where FD_S and FD_T were Feret's diameters of circles encompassing the urchins' spines and tests, respectively. Feret's diameters were obtained from the digital photographs of the aboral sides of the urchins using image analysis software, ImageJ (NIH, USA). There was no difference in the relative spine length ($F_{8,35} = 0.29$, $P = 0.9651$; mean = $32.7 \% \text{ of TD} \pm 0.3 \text{ SE}$) of *T. gratilla* among treatments at the beginning of the experiment.

Consumption rates were assessed on a daily basis for 20 days between weeks 6 and 8. The initial and final WW of *Sargassum* spp. fed to each replicate and the amount of time the seaweed was available to the sea urchins (minutes) were recorded daily. Mean change in WW of *Sargassum* spp. over 24 h recorded in a pilot trial without sea urchins ($n = 9$) in each of the three seawater exchange rates was less than $\pm 0.8 \%$, and as urchins always consumed $>50 \%$ of *Sargassum* spp. available daily, any autogenic change in WW of *Sargassum* spp. was assumed to be negligible. Consumption rates were calculated using the formula:

$$\text{Consumption} = [(S_i - S_f) / d_i] / (t_f - t_i)$$

where S_i and S_f were the initial and final WW, respectively, of the *Sargassum* spp., d_i was the initial mean WW of the urchins, and t_f and t_i were the final and initial time points, respectively.

Food conversion efficiency was calculated using the formula:

$$\text{FCE \%} = [(d_f - d_i) / S] \times 100$$

where d_i and d_f were the initial and final mean WW, respectively, of the urchins and S was the total WW of *Sargassum* spp. consumed.

The amount of CO_2 produced by *T. gratilla* during week 10 (Table S3) was estimated using b and K constant values

for *T. gratilla* established by Dy et al. (2002) under similar temperature and salinity conditions to this study, for the equation $M = KW^b$ relating body mass and metabolic rate (Webb et al. 1977).

Partitioning factors that contributed to changes in seawater parameters

Experiment 3 was conducted to determine what changes in seawater parameters (pH, A_T , DO, pCO_2 , ΩCa and nitrate) could be attributed to parameters that were likely to affect water quality, specifically the sea urchins, their faeces, the seaweed food (*Sargassum* spp.) or all of these in combination, and how these changes were influenced by light and seawater exchange rate (0.3 vs. 3.0 exchanges h^{-1}). Each treatment had five replicates, consisting of the 5 L buckets previously described. The *T. gratilla* were kept at a density of 15 individuals replicate $^{-1}$ (equivalent to 129 individuals m^{-2}). *Sargassum* spp. was stocked at 20 kg m^{-3} . Replicates with faeces only and *Sargassum* spp. only were used to determine the contribution of these factors to water chemistry. Faeces were collected from containers housing *T. gratilla* (129 individuals m^{-2} , 3.0 exchanges h^{-1} for 24 h), pooled and distributed by equal volume (0.5 L) into five replicate containers per exchange rate. Five controls without the sea urchins, faeces or *Sargassum* spp. were also used for each of the seawater exchange rates. The experiment was conducted within the previously described flow-through seawater system under a 12:12 photoperiod.

Seawater samples were collected from all replicates after 24 h during the light phase of the photoperiod (6 h after lights 'on') and an additional 12 h later (6 h after lights 'off') during the dark phase of the photoperiod. Nitrate and A_T were measured for each replicate at each sample time as previously described. Temperature, pH, DO and salinity readings were taken in each replicate at each sample time as previously described. Values for pCO_2 and ΩCa in each replicate were calculated as above.

Statistical analysis

Data for growth, survival, spine length, consumption, FCE and seawater physico-chemical parameters (nitrate, A_T , DO, pH, pCO_2 , ΩCa) were analysed using permutational analysis of variance (PERMANOVA, Anderson 2001) with temperature (experiment 1), seawater exchange rate and density (experiment 2) or light, seawater exchange rate and treatment (experiment 3) as orthogonal fixed factors. Containers were used as the level of replication. For experiment 3, there was no significant difference in levels of nitrate, A_T , DO, pH, pCO_2 or ΩCa between light and dark phases (PERMANOVA, $P > 0.25$, not shown), so replicates from the two phases were pooled (Underwood 1997).

Multiple linear regressions were used to examine the relationships between water quality parameters (temperature, nitrate, DO, salinity, pH, pCO_2 , ΩCa and A_T) and SGR, LGR, relative spine length, consumption rates and FCE in the density and seawater exchange rate experiment (experiment 2). Temperature, nitrate, DO, salinity, natural log-transformed pH, pCO_2 , ΩCa , LGR, SGR and relative spine length were averaged across weeks within each treatment to match the sampling design for A_T ($n = 10$ per treatment). LGR during week 1 was excluded from the analysis as test diameter did not increase in any treatment during this period. Three weekly mean consumption rates and FCE for each treatment were compared to DO, temperature, nitrate, salinity, natural log-transformed pH, pCO_2 , ΩCa and A_T values during weeks 6–8 of the experiment ($n = 3$ per treatment). Marginal tests were calculated for each seawater parameter, and parameters with P values > 0.01 were excluded from further analysis. For consumption (Table S1), pH was used as a proxy for DO, salinity and temperature as these parameters were highly correlated (> 0.99). Further analyses beyond marginal tests were not required for FCE (Table 3).

Linear regressions were conducted using the distance-based linear modelling (DISTLM) with distance-based redundancy analysis (dbRDA) routine of Primer 6 (Primer-E, Plymouth) with PERMANOVA $^+$ extension (version 6.1.11) software (Anderson 2001; Anderson et al. 2008; McArdle and Anderson 2001). The BEST model selection routine with AICc selection criterion, based on 9999 permutations, was used to select the five best-fitting models.

Results

Effects of temperature on growth and survival

Survival after 9 weeks was not significantly different among temperature treatments ($F_{3, 32} = 1$, $P = 1$). Temperature had a significant effect on the TD of *T. gratilla* (Fig. 1a, $F_{3, 31} = 78.09$, $P < 0.0001$, followed by post hoc pair-wise test, Fig. 1c). After 9 weeks, TD was greatest in the 26–28 °C treatment (69.9 mm, 485 % increase from initial TD), followed by the 23–25 °C treatment (59.7 mm, 401 %), then the 19–21 °C treatment (39.1 mm, 293 %), whilst the 29–31 °C treatment had the smallest increase in TD (26.2 mm, 186 %).

Temperature had a significant effect on the WW of *T. gratilla* (Fig. 1b, $F_{3, 31} = 56.95$, $P < 0.0001$, followed by post hoc pair-wise test). After 9 weeks, WW was greatest in the 26–28 °C treatment (88.4 g, 7085 % increase from initial WW), followed by the 23–25 °C treatment (54.6 g, 4117 %), then the 19–21 °C treatment (18.6 g, 1766 %),

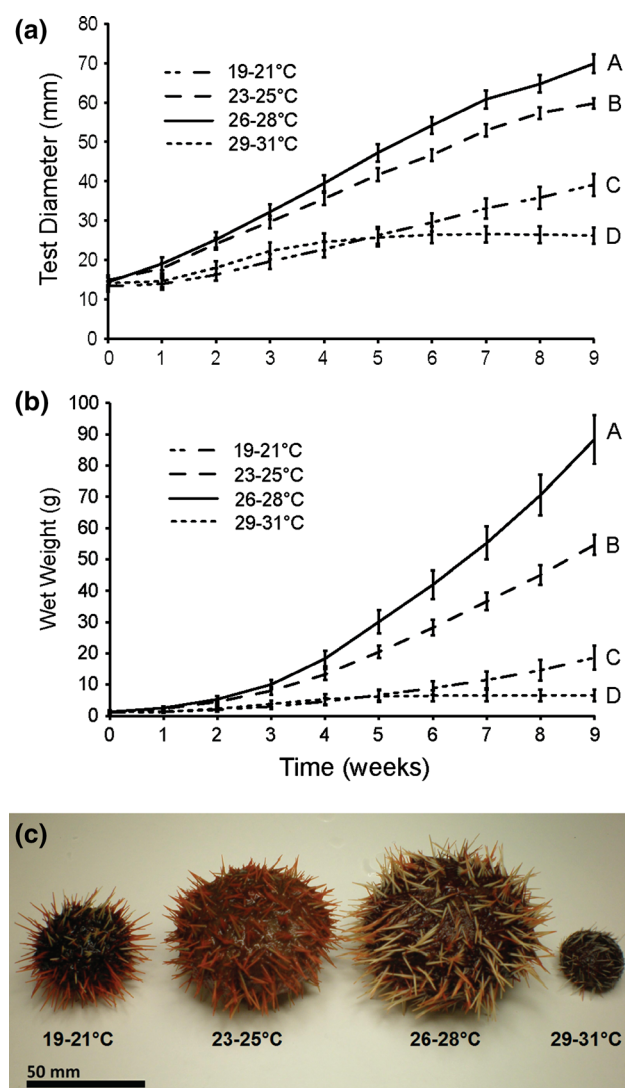


Fig. 1 Effect of temperature on the growth of *Tripneustes gratilla* over 9 weeks in a flow-through seawater system. **a** Cumulative test diameter, **b** cumulative wet weight, **c** representative mean size and colour variance at week 9. **a**, **b** Capitalised letters denote significant differences among treatments at week 9 (one-way PERMANOVA, followed by post hoc pair-wise test). Data are mean $\pm$ SE, $n = 9$ for all treatments except 29–31 °C ($n = 8$)

whilst the 29–31 °C treatment had the smallest increase in WW (6.6 g, 475 %).

Effects of density and seawater exchange rate on growth and survival

Survival was >90 % after 10 weeks and was not significantly different among treatments (Table 2). Deaths occurred only in the first 2 weeks of the experiment in four of the nine treatments.

Linear growth rate (LGR) of *T. gratilla* over the 10 weeks ranged from 2.56 to 4.60 mm week⁻¹ (Fig. 2a).

LGR was dependent on a significant interaction between seawater exchange rate and stocking density, with higher growth rates in low-density and high seawater exchange treatments and lower growth rates in high-density and low seawater exchange treatments (Fig. 2a; Table 2).

Specific growth rates (SGRs) of *T. gratilla* over the 10 weeks ranged from 22.8 to 32.0 % week⁻¹ (Fig. 2b). SGR was dependent on a significant interaction between seawater exchange rate and stocking density, with higher growth rates in low-density and high seawater exchange treatments and low growth rates in high-density and low seawater exchange treatments (Fig. 2b; Table 2).

Relative spine lengths of *T. gratilla* after 10 weeks ranged from 7 to 16 % of TD (Fig. 2c). Relative spine length was dependent on seawater exchange rate and density with no interaction between treatments (Fig. 2c; Table 2). Relative spine lengths were reduced by decreased seawater exchange rates and increased density (Table 2).

Effects of density and seawater exchange rate on consumption and FCE

Urchins consumed 6–9 % of their body mass per day (Fig. 3a). Consumption rates were dependent on a significant interaction between seawater exchange rate and stocking density (Fig. 3a; Table 2), with generally higher consumption rates at high seawater exchange rates and low densities and lower consumption rates at low seawater exchange rates and high densities.

Food conversion efficiency (FCE) ranged from 52 to 61 % (Fig. 3b). FCE was dependent on density, but not seawater exchange rate (Fig. 3b; Table 2). FCE was significantly higher in the low- and medium-density treatments than in the high-density treatment (Fig. 3b; Table 2).

Effects of density and seawater exchange rate on seawater parameters

Mean values for temperature, DO, salinity and nitrate concentration recorded in each of the treatments are reported in Table S2. TAN and nitrite were not detected (<0.00 mg L⁻¹) except during week five for a period of 72 h at a maximum of 0.06 mg L⁻¹ TAN and 0.02 mg L⁻¹ nitrite in the seawater supplied to the experiment, associated with a heavy rainfall event.

Total alkalinity (A_T) was lower in all treatments (mean 816–2112 $\mu\text{mol L}^{-1}$) compared to the ambient seawater supplied to the system (mean 2166 $\mu\text{mol L}^{-1}$). There was a general decline in A_T in all treatments during the first 6 weeks of the experiment (Fig. 4a). The decline in A_T was greatest in the low seawater exchange, high-density treatment and lowest in the high seawater exchange, low-density treatment. A_T increased in all treatments during weeks

Table 2 PERMANOVA analyses examining the effects of density and seawater exchange rate on survival, linear growth rate (LGR), specific growth rate (SGR) and relative spine length (ratio of spine length to test diameter) after 10 weeks and consumption rate and food conversion efficiency (FCE) of *Tripneustes gratilla* over 20 days in a flow-through seawater system

	Source	df	MS	F	P	Post hoc tests	
Survival	Exchange rate	2	11.94	0.39	0.6664		
	Density	2	3.13	0.10	0.9157		
	Er × De	4	53.51	1.74	0.1365		
	Residual	35	30.73				
LGR	Exchange rate	2	2.96	128.72	0.0001	low (3.0 = 1.0 > 0.3)	0.3 (low > medium > high)
	Density	2	5.56	242.11	0.0001	medium (3.0 > 1.0 > 0.3)	1.0 (low > medium > high)
	Er × De	4	0.09	3.77	0.0131	high (3.0 > 1.0 > 0.3)	3.0 (low > medium > high)
	Residual	35	0.02				
SGR	Exchange rate	2	54.18	124.61	0.0001	low (3.0 = 1.0 > 0.3)	0.3 (low > medium > high)
	Density	2	99.78	229.49	0.0001	medium (3.0 = 1.0 > 0.3)	1.0 (low > medium > high)
	Er × De	4	1.77	4.08	0.0077	high (3.0 > 1.0 > 0.3)	3.0 (low > medium > high)
	Residual	35	0.43				
	Spine length						
	Exchange rate	2	116.67	94.81	0.0001	3.0 > 1.0 > 0.3	
	Density	2	25.51	20.73	0.0001	low > medium > high	
	Er × De	4	0.40	0.33	0.8644		
Consumption	Exchange rate	2	977.55	50.74	0.0001	low (3.0 > 1.0 = 0.3)	0.3 (low > high)
	Density	2	127.92	6.64	0.0039	medium (3.0 = 1.0 > 0.3)	1.0 (low = medium = high)
	Er × De	4	94.39	4.90	0.0040	high (3.0 = 1.0 > 0.3)	3.0 (low = medium = high)
	Residual	35	19.27				
FCE	Exchange rate	2	29.77	2.41	0.1048		
	Density	2	127.56	10.35	0.0007	low = medium > high	
	Er × De	4	15.79	1.28	0.3009		
	Residual	34	12.32				

Er, exchange rate; De, density; significant differences ($P < 0.05$) are in bold; df, degrees of freedom; MS, mean square. Post hoc tests: low, medium and high correspond to low-, medium- and high-density treatments respectively. 0.3, 1.0 and 3.0 correspond to seawater exchange rate treatments (0.3, 1.0 and 3.0 exchanges h^{-1} , respectively)

7–10, but did not return to levels recorded in the treatments at the beginning of the experiment (Fig. 4a).

Seawater pH decreased in all treatments over time from ~8.12 to as low as 7.61 in the low seawater exchange, high-density treatment (Fig. 4b). pH was lower in all treatments compared to the ambient seawater supplied to the system after 3 weeks (Fig. 4b). The decline in pH was greatest in the low seawater exchange, high-density treatment and lowest in the high seawater exchange, low-density treatment.

Carbon dioxide concentrations ($p\text{CO}_2$) increased in all treatments over time from 200 to 300 μatm to as high as 1024 μatm in the medium seawater exchange, medium-density treatment (Fig. 4c). The increase in $p\text{CO}_2$ was greatest in the medium seawater exchange, high-density treatment and lowest in the high seawater exchange, low-density treatment.

$p\text{CO}_2$ was higher in all treatments compared to ambient levels in the seawater supplied to the system after 8 weeks (Fig. 4c).

Mineral saturation (ΩCa) decreased in all treatments over time by approximately 50 % (Fig. 4d). The decline in ΩCa was greatest in the low seawater exchange, high-density treatment and lowest in the high seawater exchange, low-density treatment. ΩCa was lower in all treatments compared to the ambient seawater supplied to the system after 3 weeks (Fig. 4d).

Effects of environmental parameters on growth, consumption and FCE

Marginal tests indicated $p\text{CO}_2$, A_T and ΩCa explained 19, 15 and 10 % of the variance in SGR and 47, 46 and 42 % of

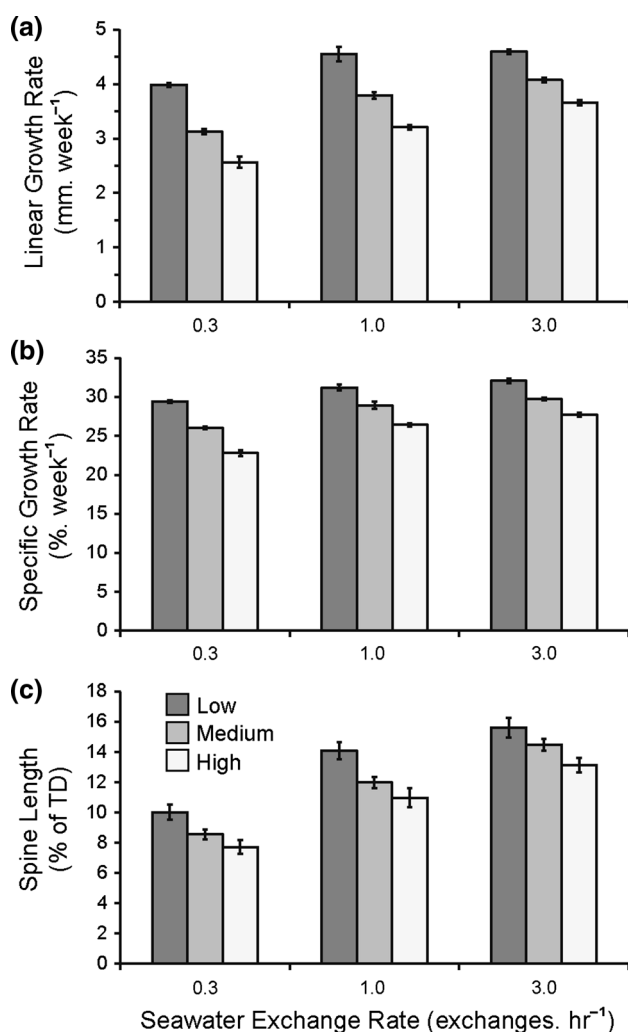


Fig. 2 Effects of density and seawater exchange rate on the growth of *Tripneustes gratilla* after 10 weeks in a flow-through seawater system. **a** Linear growth rate (LGR), **b** specific growth rate (SGR), **c** relative spine length (ratio of spine length to test diameter). Stocking density: *dark bars* low (43 individuals m⁻²); *medium bars* medium (86 individuals m⁻²); *light bars* high (129 individuals m⁻²). **a**, **b** LGR and SGR were dependent on an interaction between seawater exchange rate and density (Table 2). **c** Relative spine length was dependent on seawater exchange rate (Table 2, 0.3 < 1.0 < 3.0) and density (Table 2, low > medium > high). Data are mean $\pm$ SE, $n = 5$ except for the low-density treatment at 0.3 exchanges h⁻¹ ($n = 4$)

the variance in LGR respectively. All other seawater parameters were a poor fit ($P > 0.01$, Table 3). Likewise $p\text{CO}_2$, A_T and ΩCa explained 65, 53 and 50 % of the variance in relative spine length respectively, and all other parameters were a poor fit (Table 3). No single model of the relationship between seawater parameters and SGR, LGR or relative spine length was supported by the data as there were two or more models with similar AICc values (Table S1). Models containing all parameters (A_T , ΩCa and $p\text{CO}_2$) had similar AICc values to models containing fewer parameters (Table S1).

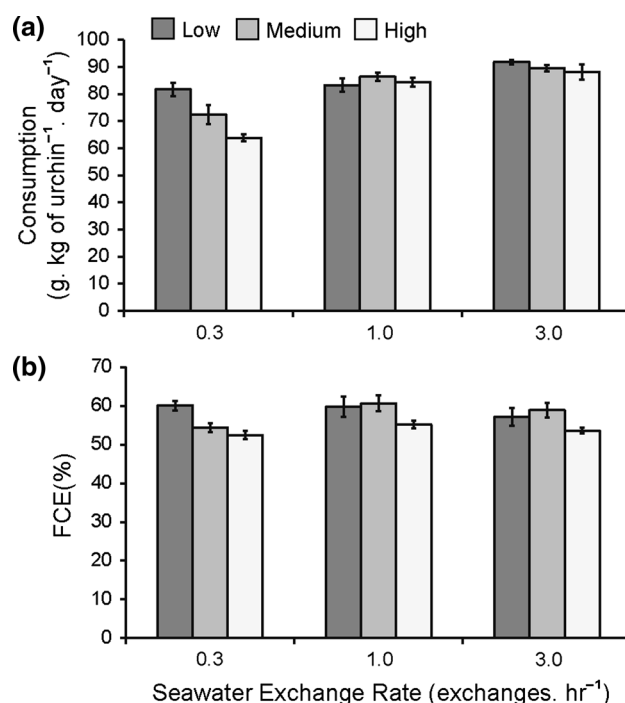


Fig. 3 Effects of density and seawater exchange rate on **a** consumption rate and **b** food conversion efficiency (FCE) of *Tripneustes gratilla* over 20 days in a flow-through seawater system. Stocking density: *dark bars* low (43 individuals m⁻²); *medium bars* medium (86 individuals m⁻²); *light bars* high (129 individuals m⁻²). **a** Consumption rates were dependent on a significant interaction between seawater exchange rate and density (Table 2). **b** FCE was dependent on density (Table 2, low = medium > high), but not seawater exchange rate (Table 2). Data are mean $\pm$ SE, $n = 5$ except for the low-density treatment at 0.3 exchanges h⁻¹ ($n = 4$)

For consumption, all seawater parameters had P values < 0.01 in marginal tests (Table 3), although A_T (51 %) and ΩCa (47 %) explained more variation than other parameters (< 40 %, Table 3). No single model of the relationship between seawater parameters and consumption rates was supported by the data as all AICc values in the top five models were similar (Table S1). Consumption rates were highly correlated with A_T , $p\text{CO}_2$, pH, salinity, DO and temperature, but ΩCa and nitrate contributed little to the models (Table S1). Marginal tests comparing seawater parameters and FCE indicated all the seawater parameters tested were a poor fit ($P > 0.09$, Table 3).

Partitioning factors that contributed to changes in seawater parameters

Mean nitrate concentrations were < 0.92 mg L⁻¹ in all treatments (*T. gratilla*, faeces, food and combined) and controls (i.e. seawater only) (Fig. 5a). Nitrate concentration was dependent on a significant interaction between seawater exchange rate and treatment (Table 4, followed

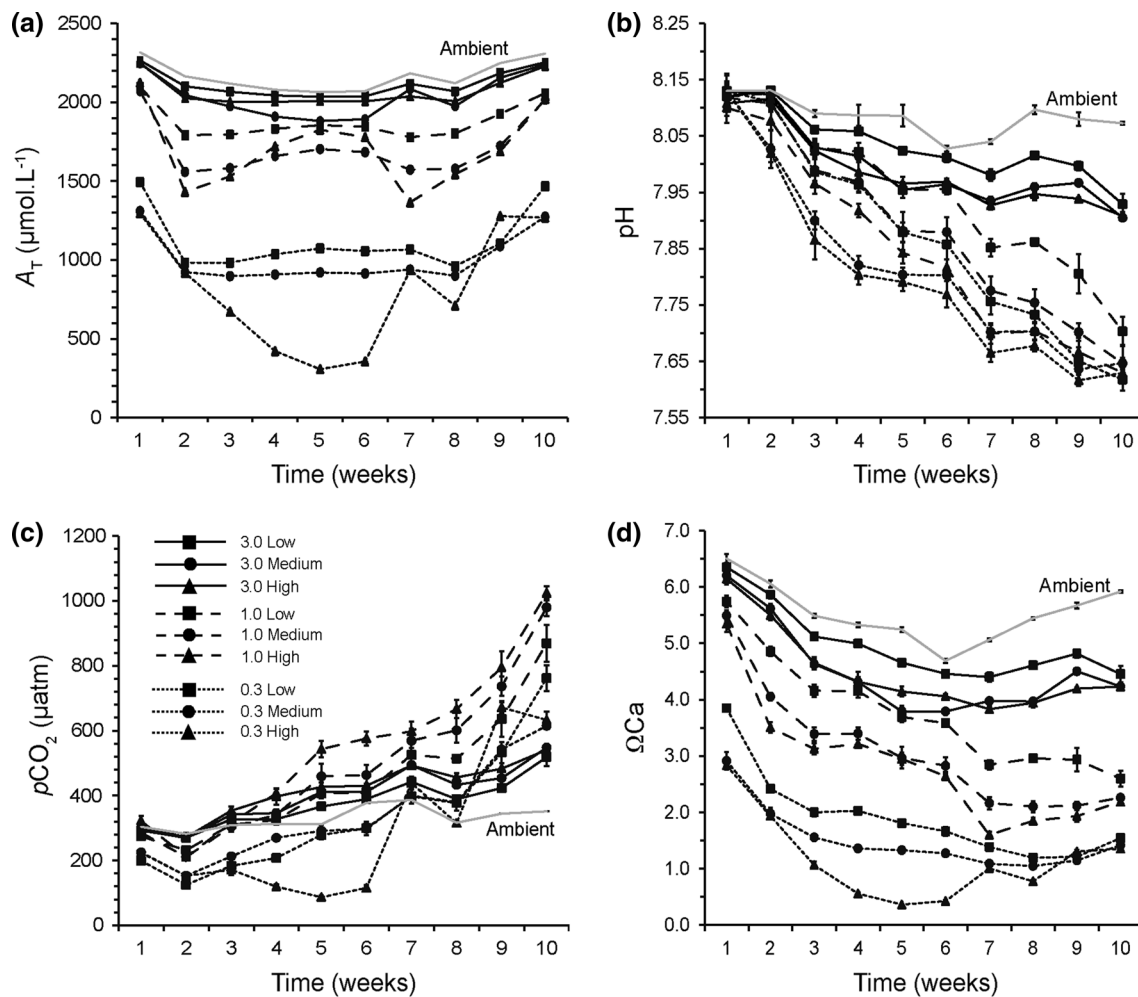


Fig. 4 Effects of seawater exchange rate and density of *Tripneustes gratilla* on water quality parameters in a flow-through seawater system over 10 weeks. **a** Total alkalinity (A_T), **b** pH, **c** dissolved carbon dioxide (pCO_2), **d** calcite saturation state (ΩCa). Seawater exchange rates: dotted lines 0.3 exchanges h^{-1} ; dashed lines 1.0 exchanges h^{-1} ; solid black lines 3.0 exchanges h^{-1} . Stocking density: squares low;

circles medium; triangles high. Ambient denotes values in the seawater supplied to the system. Data are mean $\pm$ SE. Error bars are absent for A_T as samples were pooled from replicates in each treatment. **b–d** $n = 5$, except for the low-density treatment at 0.3 exchanges h^{-1} ($n = 4$)

by post hoc pair-wise tests, Fig. 5a). At an exchange rate of 3.0 exchanges h^{-1} , nitrate concentrations were not significantly different among treatments (*T. gratilla*, faeces, food and combined) or the controls at either exchange rate. All treatments (*T. gratilla*, faeces, food and combined) at an exchange rate of 0.3 exchanges h^{-1} had significantly lower nitrate concentrations than the controls at both 0.3 and 3.0 exchanges h^{-1} and the treatments (*T. gratilla*, faeces, food and combined) at 3.0 exchanges h^{-1} . At an exchange rate of 0.3 exchanges h^{-1} , nitrate concentration was significantly lower in the combined treatment than in treatments containing only *Sargassum* spp. or *T. gratilla*.

Mean A_T was dependent on a significant interaction between seawater exchange rate and treatment (Table 4,

followed by post hoc pair-wise tests, Fig. 5b). A_T was significantly lower in the combined and *T. gratilla* only treatments than the control treatments at both exchange rates. However, at 3.0 exchanges h^{-1} , A_T was significantly lower in the *T. gratilla* only treatment than in the combined treatment. A_T was significantly higher in the faeces and *Sargassum* spp. only treatments than the control at an exchange rate of 0.3 exchanges h^{-1} .

Dissolved oxygen (DO) was dependent on treatment (*T. gratilla*, faeces, food, combined and controls), but not seawater exchange rate (Table 4, followed by post hoc pair-wise tests, Fig. 5c). DO was significantly lower in all treatments (*T. gratilla*, faeces, food and combined) compared to the controls. DO was significantly higher in the faeces and *Sargassum* spp. only treatments than in the *T. gratilla*

Table 3 Distance-based linear model (DISTLM) marginal tests for specific growth rate (SGR), linear growth rate (LGR), relative spine length (RSL), consumption rates and food conversion efficiency (FCE) of *Tripneustes gratilla* and selected environmental variables

Data set	Variable	SS (trace)	F	P	Prop.
SGR	pCO₂	2659.00	20.98	0.001	0.19
	A_T	2012.00	15.00	0.001	0.15
	ΩCa	1425.70	10.13	0.006	0.10
	Nitrate	890.04	1.73	0.015	0.06
	Salinity	234.13	1.52	0.220	0.02
	pH	177.76	1.15	0.297	0.01
	Temp.	169.94	1.10	0.282	0.01
	DO	155.67	1.00	0.325	0.01
LGR	pCO₂	60.53	63.00	0.0001	0.47
	A_T	58.46	59.03	0.0001	0.46
	ΩCa	53.64	50.64	0.0001	0.42
	Nitrate	10.75	6.43	0.012	0.08
	Salinity	1.18	0.65	0.421	0.01
	Temp.	1.06	0.59	0.453	0.01
	DO	1.04	0.57	0.446	0.01
	pH	0.97	0.54	0.466	0.01
RSL	pCO₂	2750.40	163.21	0.001	0.65
	A_T	2223.90	97.40	0.001	0.53
	ΩCa	2129.70	89.09	0.001	0.50
	Nitrate	161.17	3.48	0.053	0.04
	Salinity	107.43	2.29	0.112	0.03
	pH	76.82	1.63	0.215	0.02
	Temp.	60.31	1.27	0.276	0.01
	DO	49.82	1.05	0.324	0.01
Consumption	A_T	1.25E−3	26.46	0.0001	0.51
	ΩCa	1.14E−3	22.25	0.0001	0.47
	Salinity	9.57E−4	16.34	0.0051	0.40
	Temp.	9.27E−4	15.51	0.0065	0.38
	pH	9.20E−4	15.31	0.0069	0.38
	DO	9.16E−4	15.20	0.0074	0.38
	pCO₂	8.58E−4	13.71	0.0006	0.35
	Nitrate	6.87E−4	9.91	0.0040	0.28
FCE	Nitrate	139.12	3.04	0.093	0.11
	DO	91.58	1.92	0.183	0.07
	pH	86.38	1.80	0.208	0.07
	Temp.	84.18	1.76	0.211	0.07
	Salinity	77.27	1.60	0.223	0.06
	pCO ₂	8.18	0.16	0.687	0.00
	A _T	0.52	0.01	0.922	0.00
	ΩCa	0.16	2.45E−3	0.961	0.00

Selection procedure: BEST, selection criterion: AICc. 'Prop.' is the proportion of variation in the data set explained by the variable. Variables with $P < 0.01$ are bold

A_T, total alkalinity; ΩCa, calcite saturation state; DO, dissolved oxygen; pCO₂, dissolved carbon dioxide; Temp., temperature

only treatments, which were higher than in the combined treatments.

Seawater pH was dependent on a significant interaction between seawater exchange rate and treatment (Table 4, followed by post hoc pair-wise tests, Fig. 5d). At both exchange rates, pH was significantly lower in all treatments (*T. gratilla*, faeces, food and combined) than in the controls and was significantly higher in the faeces and *Sargassum* spp. treatments than in the *T. gratilla* only treatments. pH in *T. gratilla* only treatments was higher than in the combined treatments at an exchange rate of 3.0 exchanges h^{−1}, but not at 0.3 exchanges h^{−1}. pH was significantly higher in the *T. gratilla* and combined treatments at an exchange rate of 3.0 exchanges h^{−1} than in the same treatments at 0.3 exchanges h^{−1}, respectively.

Dissolved carbon dioxide (pCO₂) was dependent on a significant interaction between seawater exchange rate and treatment (Table 4, followed by post hoc pair-wise tests, Fig. 5e). At both exchange rates, pCO₂ was significantly higher in all treatments (*T. gratilla*, faeces, food and combined) than in the controls. Likewise in both exchange rates, pCO₂ was significantly higher in the combined treatments than the faeces and *T. gratilla* only treatments (except for the faeces only treatment at an exchange rate of 0.3 exchanges h^{−1}) which were higher than the *Sargassum* spp. only treatments. In the faeces only, *Sargassum* spp. only and combined treatments, pCO₂ was significantly higher at an exchange rate of 0.3 exchanges h^{−1} than 3.0 exchanges h^{−1}.

Mineral saturation (ΩCa) was dependent on a significant interaction between seawater exchange rate and treatment (Table 4, followed by post hoc pair-wise tests, Fig. 5f). At both exchange rates, ΩCa was significantly lower in all treatments (*T. gratilla*, faeces, food and combined) than in the controls and was significantly higher in the faeces and *Sargassum* spp. treatments than the *T. gratilla* only treatments. ΩCa in *T. gratilla* only treatments were higher than in the combined treatment at an exchange rate of 3.0 exchanges h^{−1}, but was lower at an exchange rate of 0.3 exchanges h^{−1}. ΩCa was significantly higher in the *T. gratilla* and combined treatments at an exchange rate of 3.0 exchanges h^{−1} than in the same treatments at 0.3 exchanges h^{−1}, respectively.

Discussion

At the optimal temperature identified here for *T. gratilla*, growth (weight, diameter and relative spine length), consumption rates and FCE were reduced at high densities and low seawater exchange rates. Interestingly, modelling

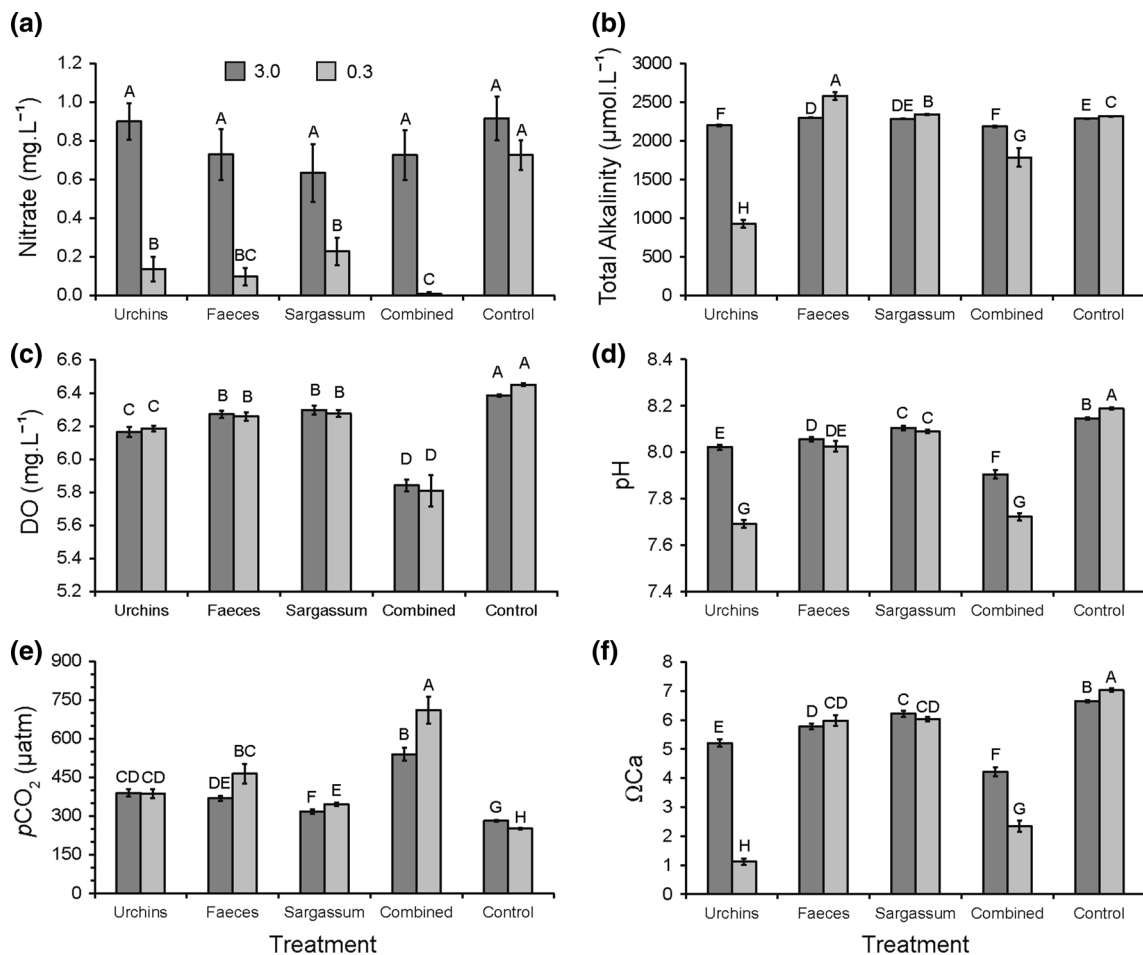


Fig. 5 Effects of seawater exchange rate and the presence of *Tripneustes gratilla*, *T. gratilla* faeces and *Sargassum* spp. on water quality parameters in a flow-through seawater system. **a** Nitrate concentration, **b** total alkalinity (A_T), **c** dissolved oxygen (DO), **d** pH, **e** dissolved carbon dioxide (pCO_2), **f** calcite saturation state (ΩCa).

Dark bars 3.0 exchanges h⁻¹; light bars 0.3 exchanges h⁻¹. Combined = faeces + *Sargassum* + urchins, control = seawater only. Bars with the same capitalised letter are not statistically different (Table 4, followed by post hoc pair-wise test). Data are mean $\pm$ SE of replicates pooled from light and dark treatments, $n = 6-10$

showed that carbonate chemistry of the culture water appeared to be a major driver for this response, with A_T , ΩCa and pCO_2 highly correlated with growth (SGR, LGR, relative spine length). Reductions in A_T and ΩCa and increases in pCO_2 in the culture water were most likely caused by the uptake of bicarbonate and release of respiratory CO_2 by the sea urchins. These results suggest that low A_T and its effects on carbonate chemistry may be a common and underappreciated determinant of productivity in intensive cultures of calcifying invertebrates and may become more important in the future as atmospheric CO_2 increases and ocean pH decreases.

The effect of temperature on growth of *T. gratilla* in this study followed the typical nonlinear pattern common to ectotherms (Huey and Kingsolver 1989). Growth increased with increasing temperature until a thermal threshold was reached at approximately 29 °C, after which there was a rapid reduction in growth. Temperatures which resulted in

the highest survival and growth rates of juvenile *T. gratilla* in this study (26–28 °C) correspond with optimal temperature ranges reported for larval growth, metamorphosis and early post-settlement of *T. gratilla* in culture (Mos et al. 2011; Rahman et al. 2009; Sheppard Brennan et al. 2010) and mean summer temperatures in much of their natural distribution (Lawrence and Agatsuma 2013).

Stocking density and seawater exchange rate did not affect survival of *T. gratilla*, which was >90 % in all treatments. This is similar to other studies that have examined the effects of stocking density on sea urchins (Christiansen and Siikavuopio 2007; Gago and Luís 2010; James 2006; Kelly 2002). Some mortalities occurred in the first 2 weeks and may have been caused by handling stress (Dale et al. 2005). Density-dependent mortality of *T. gratilla* due to a decline in water quality is reported at stocking densities much greater than those used here (250 or 750 urchins m⁻², Juinio-Menez et al. 2008).

Table 4 PERMANOVA analyses examining the effects of seawater exchange rate and presence of *Sargassum* spp., *Tripneustes gratilla* and *T. gratilla* faeces on nitrate concentration, total alkalinity (A_T), dissolved oxygen (DO), pH, carbon dioxide (pCO_2) and calcite saturation state (ΩCa) in a flow-through seawater system

	Source	df	MS	F	P
Nitrate	Exchange rate	1	0.34	78.02	0.0001
	Treatment	4	0.03	7.05	0.0002
	Er $\times$ Tr	4	0.01	3.10	0.0199
	Residual	80	4.30E-3		
A_T	Exchange rate	1	1.66E6	86.22	0.0001
	Treatment	4	2.38E6	123.35	0.0001
	Er $\times$ Tr	4	1.80E6	93.50	0.0001
	Residual	88	1.93E4		
DO	Exchange rate	1	2.09E-4	0.02	0.8931
	Treatment	4	0.82	69.84	0.0001
	Er $\times$ Tr	4	5.65E-3	0.48	0.7448
	Residual	79	1.17E-2		
pH	Exchange rate	1	0.26	158.38	0.0001
	Treatment	4	0.46	280.99	0.0001
	Er $\times$ Tr	4	0.11	67.65	0.0001
	Residual	88	1.65E-3		
pCO_2	Exchange rate	1	6.28E4	11.69	0.0007
	Treatment	4	3.63E5	67.59	0.0001
	Er $\times$ Tr	4	3.21E4	5.97	0.0001
	Residual	84	5.37E3		
ΩCa	Exchange rate	1	28.84	202.67	0.0001
	Treatment	4	55.15	387.62	0.0001
	Er $\times$ Tr	4	15.82	111.2	0.0001
	Residual	84	0.14		

The post hoc tests for the interactions between exchange rate and treatment for all physico-chemical parameters are presented in Fig. 5

Er, exchange rate; De, density; significant differences ($P < 0.05$) are in bold; df, degrees of freedom; MS, mean square

Growth rates observed across all treatments were much higher than reported for any other juvenile urchin in culture (Kelly 2002; Siikavuopio and James 2011). Growth of *T. gratilla* was strongly influenced by stocking density and water exchange. SGR and LGR were 41 and 79 % higher, respectively, in the high seawater exchange, low-density treatment compared to the low seawater exchange, high-density treatment, and relative spine length showed a similar pattern. Growth of other juvenile sea urchins is also reduced at high densities (Kelly 2002; Siikavuopio and James 2011). In contrast, Juinio-Menez et al. (2008) found that growth of adult *T. gratilla* in sea cages was not affected by stocking density, although growth rates were >2.5 times lower than for the juveniles in this study.

We measured several physico-chemical parameters (pH, A_T , DO, salinity, temperature, TAN, nitrite and nitrate) to identify factors that correlated with variation in growth

among our treatments. The treatments (experiment 2) with the lowest growth rates had low A_T and ΩCa , and high pCO_2 . Confirming this observation, model selection using a modified Akaike information criterion (AICc) found that growth (SGR, LGR, relative spine length) was best correlated with A_T , ΩCa and pCO_2 (Table 3, Table S1). Surprisingly, variation in other parameters that are usually singled out as depressing growth in high intensity aquaculture, such as DO and nitrogenous wastes, were poorly correlated with growth. Whilst correlations are not necessarily indicative of causation, our models provide strong evidence that seawater carbonate chemistry was likely to be the primary cause of differences in the growth of *T. gratilla* among treatments. It is important to highlight that seawater carbonate chemistry was not directly manipulated. pCO_2 and ΩCa covary and are dependent upon A_T . This may explain why our models were unable to separate these parameters (Table S1). Studies that independently manipulate A_T , ΩCa and pCO_2 may provide greater insight into how particular components of seawater carbonate chemistry affect growth in culture systems.

High pCO_2 reduces growth in sea urchins by inducing physiological hypercapnia or disrupting internal acid-base balance (Hofmann and Todgham 2010; Portner 2012; Stumpp et al. 2012). Studies that have tested the effects of high pCO_2 on sea urchins, including *T. gratilla*, have found impaired growth and development of larvae (Albright et al. 2012; Byrne et al. 2011, 2013; Doo et al. 2012; Sheppard Brennand et al. 2010; Stumpp et al. 2012) and reductions in growth, consumption rates and FCE of juveniles and adults (Albright et al. 2012; Byrne et al. 2014; Grosjean et al. 1998; Shirayama and Thornton 2005; Siikavuopio et al. 2007c; Stumpp et al. 2012; Wolfe et al. 2013a, b). Like the present study, other species of dissolved inorganic carbon would have covaried with pCO_2 in all of these studies. There is, however, growing consensus that CO_2 is the main species of dissolved inorganic carbon responsible for stunting the growth of sea urchins (Byrne et al. 2013; Dubois 2014; Holtmann et al. 2013; Stumpp et al. 2012).

It is unlikely that A_T had any direct effect on growth. Rather, low A_T reduces the buffering capacity of seawater and therefore decreases the relative ΩCa for a given pCO_2 (see Fig. 5). Reduced ΩCa may affect growth rates of calcifying organisms by increasing the energy required to precipitate carbonate structures, reducing energy available for growth, whilst at $\Omega Ca < 1$, carbonate structures dissolve and calcification is impaired (Doney et al. 2009; Holtmann et al. 2013; Miles et al. 2007; Siikavuopio et al. 2007c). *T. gratilla*, like other tropical sea urchins, produces high Mg^{2+} calcite which theoretically is more soluble compared to low Mg^{2+} calcite found in sea urchins at high latitudes (Dubois 2014). This may render it particularly vulnerable to low carbonate saturations. We found that, for example,

spines were shortest in the low seawater exchange, high-density treatment where ΩCa was undersaturated for a period of 3 weeks. This may have been because the spines grew more slowly or that they were weaker and broke more easily, similar to *Strongylocentrotus droebachiensis* when grown in high $p\text{CO}_2$ (Holtmann et al. 2013; Siikavuopio et al. 2007c). Similarly, Byrne et al. (2014) found that *T. gratilla* grown at low pH had thinner and weaker tests.

It is unclear which seawater parameters were primarily responsible for driving differences in consumption rates. Consumption rates were highly correlated with A_T , $p\text{CO}_2$, pH, salinity, DO and temperature in our models (Table S1). Some sea urchins have lower consumption rates under high $p\text{CO}_2$ (Siikavuopio et al. 2007c) and low or high salinities (Basuyaux et al. 1998), but consumption rates are unaffected by variation in temperature (Klinger et al. 1986; Lawrence et al. 2009) or stocking density (Siikavuopio et al. 2007a). In our study, pH, salinity, DO and temperature were highly correlated (>0.99) in marginal tests and the greatest variation in these parameters occurred among seawater exchange rates. It appears that variation in consumption rates were driven by the accumulative impact of multiple factors on overall water quality rather than changes in individual seawater parameters per se.

The sea urchins rather than their food or faeces likely played the substantial role in altering A_T and other components of the carbonate chemistry of the culture water. Release of respiratory CO_2 was likely responsible for the elevated $p\text{CO}_2$ in treatments. Modelling using literature-derived constants showed the amount of CO_2 produced by the urchins through respiration (Table S3) would have been sufficient to generate the $p\text{CO}_2$ measured in the treatments. Supporting this, in experiment 3, large changes in $p\text{CO}_2$ were only recorded when the sea urchins were actively eating and respiring. A_T was significantly reduced in treatments containing only sea urchins, indicating that this was most likely caused by the uptake of bicarbonates for calcification. Decomposition and photosynthesis had limited effects with only small changes in carbonate chemistry in treatments containing only seaweed or faeces (experiment 3).

There is growing concern about the effect of anthropogenically driven changes in seawater carbonate chemistry on calcifying organisms. Sea urchins and many other calcifying marine invertebrates exhibit reduced growth under $p\text{CO}_2$ predicted for the turn of the century and early next century (e.g. Byrne et al. 2011, 2013; Gazeau et al. 2013). It is difficult to relate the results of this study to predictions for ocean acidification as the carbonate chemistry of the culture treatments progressively changed as the urchins grew. However, carbonate chemistry of the treatments (experiment 2) were in the ranges predicted for near future marine conditions (~ 0.3 – 0.4 pH unit decrease below ambient, IPCC 2013). Under these conditions, *T. gratilla*

exhibited reduced growth. This result highlights that as well as being influenced by seawater carbonate chemistry, calcifying organisms directly influence the carbonate chemistry of their environment. This needs to be considered in ocean acidification experiments investigating the influence of anthropogenic CO_2 . The influence of biogenic CO_2 and community metabolism in general is also important for modelling the impact of predicted ocean acidification in natural habitats (Anthony et al. 2011; Hurd et al. 2011).

This study is the first to our knowledge to experimentally demonstrate that calcifying organisms in intensive culture systems may experience retarded growth due to acidification of their culture water caused in part by reduction in A_T during calcification and release of respiratory CO_2 , although it has been recognised as a potential problem for some time (Grosjean et al. 1998). Low pH in high-density cultures of other calcifying organisms has been documented, indicating that this might be a problem for species other than sea urchins (e.g. Naylor et al. 2011). One of the implications of this study is that predicted increases in atmospheric CO_2 may have a disproportionate impact on the aquaculture of calcified invertebrates as increases in CO_2 will have comparatively greater effect on carbonate chemistry where buffering capacity of culture water has been reduced by uptake of carbonates. 'Culture acidification' could be countered by carbonate supplementation or other means of acidity regulation that maintain buffering capacity, or through high seawater exchange rates or low stocking densities. However, these measures would need to be balanced against associated increased production costs.

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